Intermediate Elective

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Set up

Packages

```
# reads in tidyverse package from library
library(tidyverse)
# reads in janitor package from library
library(janitor)
# reads in here package from library
library(here)
# reads in naniar package from library
library(naniar)
# reads in patchwork package from library
library(patchwork)
# reads in viridis package from library
library(viridis)
# reads in scales package from library
library(scales)
```

Data

```
# read in weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))
# read in water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))
# read in water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

```
# reads in eBird data as birds
birds <- read_csv(here("data", "birds.csv"))
# read in Venco water data
venoco_water <- read_csv(here("data","venoco-water.csv"))
```

1. Explain your inspiration

> **Note**
>
> In 1-2 sentences each, describe:
> which visualization you chose for your inspiration why it makes sense for your dataset of choice which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.)

2. Plan your figure

> **Note**
>
> Sketch what your figure would look like following your inspiration.
> Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

> **Note**
>
> Clean/wrangle your data as needed.
> Then, code the figure you decided to create.
> Make sure all your code is annotated, and that the output is showing.
> At the end of this section, include code to save your final figure as a .jpg or .png file.

4. Describe your process

> **Note**
>
> In 1-3 sentences each, describe:
> your process of using someone else's code to create a visualization challenges you encountered as you made your visualization how your visualization differs from visualizations you made in class, for assignments, or for a previous elective